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Implementation of Smart Library in Improving Services at the Gowa Regency Library and Archives Office

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Abstract : *The rapid evolution of information technology has prompted the Gowa Regency Library and Archives Office to innovate through the strategic implementation of a Smart Library service system. This qualitative descriptive study aims to thoroughly analyze how the Smart Library implementation enhances overall public service quality. Data collection was systematically conducted through observations, semi-structured interviews, and document analysis involving key stakeholders, including library staff, digital service managers, service coordinators, and library patrons. The collected data were analyzed using the interactive model by Miles and Huberman, while the program implementation was comprehensively evaluated based on David C. Korten's theoretical framework, which examines three core dimensions: the program itself, the implementing organization, and the target beneficiaries. The findings indicate that the Smart Library initiative has been executed effectively, significantly contributing to the structural enhancement of library service delivery. The integration of digital platforms—specifically the iGowa application, the Online Public Access Catalog (OPAC), and official social media channels—has successfully facilitated seamless access to information, expedited routine service processes, and increased active public participation in utilizing technology-driven library resources. However, several operational challenges remain evident. The full potential of the program is currently constrained by unstable internet network connectivity, limited supporting physical and technical infrastructure, and varying levels of digital literacy among the local community. To ensure the long-term sustainability and optimal performance of the Smart Library system, strategic interventions are necessary. These include strengthening technological infrastructure, expanding human resource capacity through targeted professional training, and conducting continuous digital literacy education programs for the public.*

Keywords: *Implementation, Smart Library, Public Service, Digital Library, Gowa Regency.*

1. Introduction

The development of information and communication technology (ICT) has brought major changes in various fields, including library management. If in the past libraries only functioned as a place to store and borrow books,

now their role has developed into a digital information service center that emphasizes efficiency, ease of access, and global connectivity (Bagchi, 2022) This change is in line with the concept of smart libraries, which are libraries that utilize digital technology, the Internet of Things (IoT), and automation systems to improve the quality of services for their

Globally, the implementation of smart libraries has proven to be able to support open access to knowledge, speed up services, and improve user experience. Research conducted by (Taher et al., 2019) confirms that the use of open source digital library software such as DSpace and Greenstone is able to improve collection management systems, facilitate information search, and expand the distribution of knowledge more effectively. This shows that technology-based libraries have an important role as knowledge centers that are adaptive to the development of people's needs in the digital era.

The concept of smart library represents the transformation of library functions from just a collection storage place to a digitally integrated information service center. The implementation of smart libraries includes the development of network infrastructure, the implementation of an automated library system (Integrated Library System / ILS), application-based user services (mobile/web), and automatic circulation mechanisms (e.g. RFID or integrated barcode), as well as digital literacy improvement programs for users and librarians. This approach not only speeds up and simplifies technical services such as cataloging, searching, and borrowing processes but also demands governance restructuring, improving human resource competencies, and data management policies that ensure accessibility, reliability, and security of services (Nabiolahe & Ghalavand, 2024)

In Indonesia, regional libraries are required to modernize to remain relevant and able to meet the needs of the community. The concept of smart libraries is one of the innovative strategies in improving access to

information, transparency of services, and supporting digital literacy. According to (Nabiolahi & Ghalavand, 2024) the success of the implementation of smart libraries is greatly influenced by the availability of technological infrastructure, human resource competence, and local government policy support that encourages digital-based public services.

Therefore, the performance of library services is very important to assess how far the services provided by the library are in meeting the expectations and satisfaction of the community. The library is the focal point of the scientific office because the library is through the accessibility of the potential of work assets (Sayogo & Yuli, 2021)

This condition is very much in accordance with the role of the Gowa Regency Library and Archives Office, which is responsible for providing information and literacy services for the community. The implementation of smart library-based programs is expected to be able to improve service quality, both in terms of efficiency, service speed, and user satisfaction. In addition, this step is also in line with the local government's vision in building digital-based governance as part of the development of Smart City in Gowa Regency.

The implementation of smart libraries in Gowa Regency began in stages in 2021 by the Gowa Regency Library and Archives Office. That year was the beginning of the digitization of library services, marked by the launch of the online digital library application (iGowa) which allows users to access collections, make loans, and extend the borrowing period online. The presence of this service signifies a shift in the function of regional libraries from just a place to store physical collections to a digital information service center that is more responsive to the needs of today's society. The initial implementation of smart libraries is also in line with the vision of Smart Library-based development in Gowa Regency, making libraries not only function to improve public literacy, but also support local government initiatives in creating a technology-based public service

management system. Although it is still faced with infrastructure challenges, especially in rural areas with limited internet networks, the implementation of smart libraries in 2021 has opened up opportunities for the development of wider digital library services in the following years. (He, 2020)

Based on the statistics of visits from the Gowa Regency Regional Public Library Service for the period 2022 to 2024, it can be depicted in the following diagram:

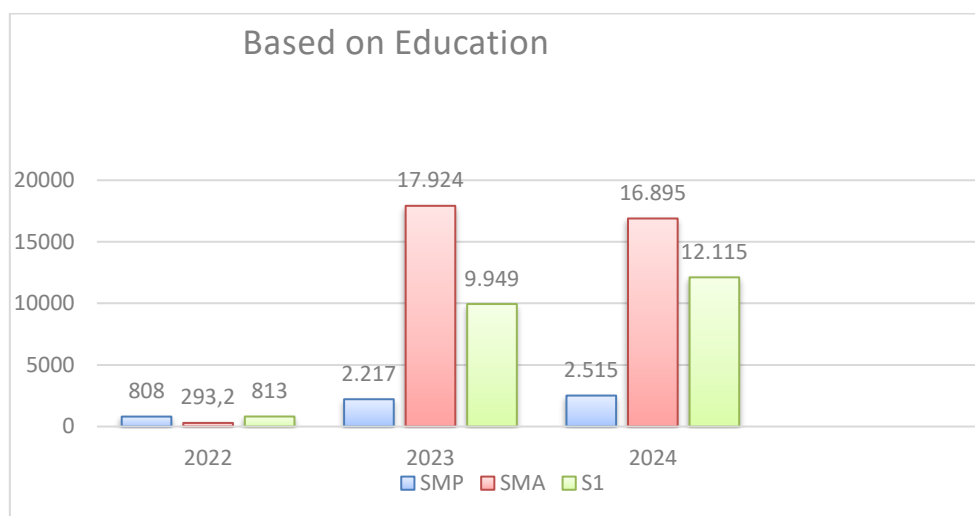


Figure 1. Data by Education

Source : Gowa Regency Library and Archives Office

Based on the image above, in the aspect of education level, respondents

The bar chart shows the distribution of respondents based on education level in 2022–2024 which consists of three categories, namely junior high school, high school, and S1.

In 2022, the number of respondents with junior high school education was recorded at 808 people, high school 293 people, and S1 as many as 813 people. The data shows that respondents with S1 education have the most number compared to other categories, although the difference is not too far from respondents with a junior high school education. Meanwhile, respondents with high school education were the group with the least number.

Entering 2023, the number of respondents in all education categories has increased. Respondents with a junior high school education increased to 2,217 people, respondents with a high school education increased significantly to 17,924 people, and respondents with a S1 education reached 9,949 people. This year, the high school category is the most dominant group with a much higher number than other education categories.

In 2024, the number of respondents with a junior high school education will again increase to 2,515 people. S1 educated respondents also increased to 12,115 people. Meanwhile, the number of respondents with high school education decreased slightly to 16,895 people, but remained the category with the highest number of respondents compared to junior high school and S1.

Based on these data, it can be concluded that the respondents' education level has experienced significant development during the 2022–2024 period. Respondents with a high school education level dominated in 2023 and 2024, while respondents with a high school education showed a consistent upward trend every year. The number of respondents with a junior high school education also increased, although it was still below the number of respondents with high school and S1 education. This condition shows that the majority of respondents in the study have a background in secondary education to higher education, which can affect their understanding and perception of the research object.

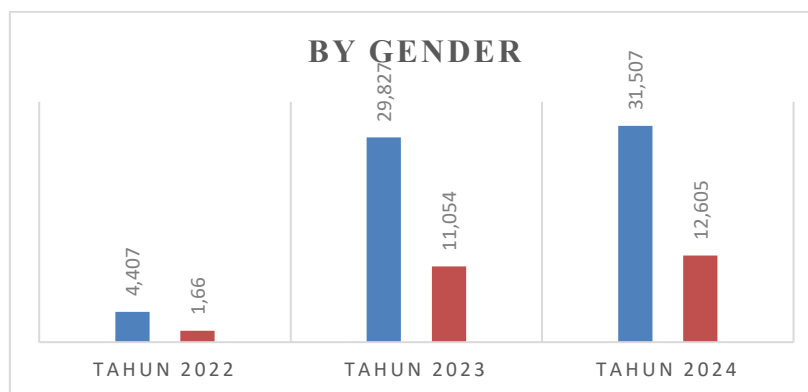


Figure 2. By Gender

Source : Gowa Regency Library and Archives Office

The bar chart shows the distribution of respondents by gender in 2022–2024 consisting of female and male categories.

In 2022, the number of female respondents was recorded at 4,407 people, while the number of male respondents was 1,660 people. The data shows that the number of female respondents is higher than that of male respondents. The considerable difference in number indicates that women's participation in this study was more dominant at the beginning of the observation period.

Entering 2023, the number of respondents in both categories has increased significantly. Female respondents increased to 29,827 people, while male respondents reached 11,054 people. Although both groups experienced an increase, the number of female respondents remained higher than that of men. This shows that the dominance of female respondents will continue in 2023.

In 2024, the number of female respondents will again increase to 31,507 people, while the number of male respondents will increase to 12,605 people. This increase shows a growth in the number of respondents in both categories compared to the previous year. Nevertheless, female respondents remained the most dominant group during the observation period.

Based on this data, it can be concluded that the number of respondents by gender has increased from 2022 to 2024. Female respondents consistently had a larger number than male respondents in each year of observation. This condition shows that the characteristics of the respondents in this study are dominated by women, so that it can give an idea that women's participation in the research object is higher than that of men during that period.

The existence of the Smart Library, the library not only functions as a place to store books, but also becomes a center for learning, innovation, and collaboration in line with the vision of a smart city. In Makassar, the success of Smart Libraries can be one of the indicators of the success of the Smart City program in improving the quality of life of the community through literacy and access to knowledge.

2. Method

This research uses a qualitative method with a descriptive research type. The qualitative approach was chosen because this study aims to gain an in-depth understanding of the Smart Library implementation process in improving the quality of services at the Gowa Regency Library and Archives Office. Through this approach, researchers can describe factually the implementation of the program, the role of the implementers, the benefits felt by the community, as well as the supporting and inhibiting factors in the implementation of Smart Library.

This research was carried out at the Gowa Regency Library and Archives Office. The selection of the research location was based on the consideration that the agency has implemented technology-based service innovations through the Smart Library program as an effort to improve access and quality of library services to the community. The Smart Library program is realized through various digital services, such as the iGowa application, Online Public Access Catalog (OPAC), and the use of social media as a means of disseminating information and services to library users.

The data sources in this study consist of primary data and secondary data. Primary data was obtained through direct interviews with informants who had involvement and knowledge related to the implementation of the Smart Library. Research informants are determined using purposive sampling techniques, which are the selection of informants based on certain considerations according to research needs. The informants in this

study consist of library staff, digital service managers, service coordinators, and library visitors. Meanwhile, secondary data was obtained through documents, archives, activity reports, and various literature related to the implementation of digital-based library services. (Miles et al., 2013)

Data collection techniques are carried out through observation, interviews, and documentation. Observation was carried out by directly observing the implementation of Smart Library services and available digital service activities. Interviews were conducted in depth to obtain information about the implementation of the program, the suitability between the program and user needs, and the obstacles faced in the implementation of Smart Library. Documentation is used to complete research data through various supporting documents related to the Smart Library program.

The data analysis in this study uses the Miles, Huberman, and Saldana data analysis model which consists of three stages, namely data condensation, data presentation, and conclusion drawing and verification. Data condensation is carried out by selecting and focusing information that is relevant to the research objective. Furthermore, the data is presented in the form of a descriptive description to facilitate the interpretation process of the phenomenon being studied. The last stage is carried out by drawing conclusions based on the findings of the research that has been analyzed. (Sugiyono, 2000)

To ensure the validity of the data, this study uses triangulation techniques which include source triangulation and triangulation techniques. Source triangulation is carried out by comparing information from various informants, while technical triangulation is carried out by comparing the results of observations, interviews, and documentation. The analysis of Smart Library implementation in this study uses David C. Korten's program implementation theory which includes three main aspects, namely program

suitability, implementation ability, and suitability with target groups. (Korten, 1987)

3. Results and discussion

The implementation of Smart Library at the Gowa Regency Library and Archives Office is a form of information technology-based public service innovation that aims to improve the quality of library services through the digitization of the service system. This program is realized through the use of the iGowa application, Online Public Access Catalog (OPAC), digitization of library collections, and the provision of various electronic-based information services. These innovations are expected to be able to provide convenience for the public to obtain access to information more quickly, effectively, and efficiently. (Farkhari et al., 2024)

Based on the results of observations, interviews, and documentation, the implementation of Smart Library was analyzed using the implementation theory proposed by (Korten, 1987). This theory emphasizes that the success of the implementation of a program is influenced by the compatibility between the implementing organization, the program being run, and the target group. These three elements are interrelated in determining the effectiveness of program implementation so that if one of the elements has not run optimally, then the program goals will be difficult to achieve.

In this study, the analysis is focused on these three indicators to find out the extent to which the implementation of Smart Library has been able to improve the quality of services at the Gowa Regency Library and Archives Office. The analysis was carried out by integrating the results of interviews with informants, the results of field observations, documentation data, and associated with relevant implementation theories and previous research. (Bagchi, 2022)

Implementing Elements

The success of the implementation of a program is not only determined by the quality of the program designed, but also greatly influenced by the capacity of the organization responsible for implementing it. In the perspective of policy implementation proposed by (Korten, 1987), the implementing organization is one of the important elements that determine the success of the implementation of a program. The implementing organization is required to have a compatibility between the tasks carried out and the organization's capabilities, both in terms of human resources, organizational structure, division of tasks, coordination, and the availability of supporting facilities and infrastructure. If the implementing organization has adequate capacity, then the program's goals will be easier to achieve effectively and sustainably.

In the context of this research, the Gowa Regency Library and Archives Office plays a role as an organization responsible for implementing the Smart Library Program as an innovation in information technology-based library services. The implementation of the program not only requires the availability of digital applications, but also requires organizational readiness to manage changes in the service system from conventional patterns to digital-based services. Therefore, organizational readiness is a factor that greatly determines the success of the implementation of Smart Library, considering that the entire service process, starting from digital collection management, circulation services, iGowa application management, to assistance to users, depends on the ability of the implementing organization. (Duda et al., 2021)

Based on the results of observations during the research, the implementation of Smart Library involves various elements in the organization, including librarians, digital service staff, service coordinators, and other support personnel who have duties and responsibilities in accordance with their respective fields. Each element

has a complementary role so that the service process can take place in an integrated manner. A clear division of duties is one of the factors that supports the smooth implementation of the program, because each employee understands his or her functions and responsibilities in providing services to the community.

In addition, organizational readiness is also reflected in the availability of organizational structure, the number of human resources, and the composition of employees who support the implementation of library services. This information is important to describe the organization's capacity to implement Smart Libraries. The more adequate the resources that the organization has, the greater the chance of successful program implementation. On the other hand, limited human resources and infrastructure have the potential to become obstacles in the implementation of digital-based services.

The characteristics of the implementing organization which include the number of employees, the composition of employees by gender, and the organizational structure of the Gowa Regency Library and Archives Service are presented in Table 1.

Table 1. Characteristics of Smart Library Implementing Organizations

Yes	Components	Results
1	Number of Employees	25 people
2	Employment Status	Civil servants: 15 people; PPPK: 4 people; Outsourcing: 6 people
3	Composition of Employees	Males: 8 people (32%); Female: 17 (68%)
4	Organizational Structure	Head of Service, Secretariat, Library Division, Archives Division, Functional Position Group, and other supporting units.
5	Smart Library Implementer	Librarians, Digital Service Staff, Service Coordinators, and support staff.

Source: Processed from the Gowa Regency Library and Archives Office (2026).

Based on Table 1, the Gowa Regency Library and Archives Service is supported by 25 employees consisting of 15 Civil Servants (PNS), 4

Government Employees with Employment Agreements (PPPK), and 6 outsourced personnel. The composition shows that the implementation of the Smart Library Program is not only supported by state civil servants, but also by support personnel who assist library service operations. In terms of employee characteristics, the organization is dominated by female employees as many as 68% (17 people), while male employees as many as 32% (8 people). In addition, the organizational structure that has been established shows a clear division of tasks between fields so that each unit has responsibilities according to their respective functions in supporting the implementation of Smart Library.

The results of the interview with the Young Expert Librarian Staff explained that the implementation of the Smart Library is supported by a clear division of tasks, ranging from the management of the iGowa application, updating digital collections, services to users, to maintaining the library information system. This condition shows that the organization already has a working mechanism that supports the implementation of the program.

Based on the results of observations, officers were able to provide assistance to the community in using OPAC and the iGowa application. Employees are also active in helping visitors who have difficulty accessing digital services. This responsive attitude is one of the factors that improve service quality.

However, the research also found several obstacles, namely the limited number of personnel who have competence in the field of information technology and the need to increase capacity through continuous training. In addition, some technological devices need updates so that digital services can run more optimally.

According to (Korten, 1987), the implementing organization must have the ability to comply with the demands of the program so that the implementation can run effectively. Based on the findings of the study, the

existence of 25 employees with a clear organizational structure has supported the implementation of the Smart Library. However, the relatively limited number of human resources compared to the demands of digital service transformation causes organizations to still face obstacles in terms of mastering information technology and maintaining digital infrastructure. Therefore, in addition to supporting the organizational structure, improving human resource competencies and strengthening technological facilities are important factors to ensure the sustainability of Smart Library implementation. These findings are in line with the research of (Duda et al., 2021) who stated that the success of Smart Libraries is greatly influenced by organizational readiness, human resource competence, and technological infrastructure support.

Programs Run

The program is the second element in the policy implementation model of (Korten, 1987). According to Korten, a program can be said to be successful if the program design is in accordance with the needs of the target group and is able to be implemented by the implementing organization. Thus, the success of implementation is not only measured by the existence of the program, but also by how the program is run through the provision of services, the use of technology, and innovations that are able to provide benefits to the community.

In this study, the Smart Library Program at the Gowa Regency Library and Archives Office is a public service innovation that utilizes information technology to improve the quality of library services. This program is realized through the provision of library collections that are constantly updated, digital library services based on the iGowa application, a collection search system using the Online Public Access Catalog (OPAC), the use of social media as a means of disseminating information, and setting service hours that provide convenience for the public to access library services.

As a form of support for the implementation of the Smart Library Program, the Gowa Regency Library and Archives Office continues to develop library collections so that the information needs of the community can be met. These conditions can be seen in Table 2.

Table 2. Number of Library Collections of the Gowa Regency Library and Archives Office

No	Year	Number of books	Number of titles
1.	2021	138	91
2.	2022	4.846	1.012
3.	2023	3.665	996
4.	2024	1.178	418
5.	2025	551	303
Total		2.820	10.378

Source: Gowa Regency Library and Archives Service (2025).

Based on Table 2, the available library collection consists of various types of library materials that are used as a source of information for the community. The periodic addition of collections shows the organization's commitment to meeting users' information needs. The availability of diverse collections is also a supporting factor for the success of Smart Libraries because digital services do not only depend on applications, but also on the quality and completeness of available library materials.

In addition to being supported by library collections, the implementation of Smart Library is also supported by service hours that provide opportunities for the public to make optimal use of library services.

Table 3. Service Hours of the Gowa Regency Library and Archives Service

NO	Remarks	Time
1.	Monday - Friday	08.00 - 16.00
2.	Saturday - Sunday	Holidays

Source: Gowa Regency Library and Archives Service (2025).

Based on the table above, it can be seen that the service time at the Gowa Regency Library and Archives Office takes place from Monday to

Friday, from 08.00 to 16.00 WITA. At this hour, the public can take advantage of various services provided by the agency as needed. Meanwhile, on Saturday and Sunday, the service does not operate because it is a holiday. Thus, it can be concluded that services are only carried out on weekdays in accordance with the provisions that apply in the agency.

Table 4. Gowa Regency Library and Archives Office Instagram Social Media Impressions

No	Month	Pengunjung
1.	November	15,432 Tayangan
2.	December	18,017 Leaning
3.	January	20,178 Tayangan
4.	February	30,309 Tayangan
5.	March	55,219 Tayangan

Source: Gowa Regency Library and Archives Service (2025).

Based on data on the number of impressions on Instagram, there is an increase from month to month. In November, the number of impressions was 15,432, then increased in December to 18,017 impressions. This shows that there has been an increase in the reach of information to the public.

Furthermore, in January the number of impressions rose again to 20,178, and there was an even greater increase in February of 30,309 impressions. The peak of the increase occurred in March with the number of impressions reaching 55,219. This significant increase shows that the content presented is increasingly able to attract users' attention.

Overall, the data shows a consistent upward trend. This can be interpreted that the use of Instagram as a medium for disseminating information has been running quite well and effectively in reaching the wider community.

Target Group

The target group is the third element in the policy implementation model of (Korten, 1987). The success of the implementation of a program is

determined by the suitability between the goals of the program and the needs of the target group that is the beneficiary. In the context of this research, the target group of the Smart Library Program includes students, students, teachers, state civil servants (ASN), and the general public who utilize the services of the Gowa Regency Library and Archives Service. Through the use of information technology, Smart Library is expected to be able to increase public access to information sources faster, easier, and more efficiently so that the quality of library services can be further improved.

To find out the extent to which the Smart Library Program has been utilized by the target group, it can be seen from the development of the number of library visitors after the implementation of the program as presented in Table 5.

Table 5. Number of Library Visitors in 2022–2025

No.	Year	Number of Visitors
1.	2022	6.067
2.	2023	40.881
3.	2024	44.112
4.	2025	48.985

Source: Gowa Regency Library and Archives Service (2025)

Based on Table 5, the number of library visitors shows an increasing trend every year after the implementation of the Smart Library Program. In 2022, the number of visitors was recorded at 6,067 people, then increased significantly to 40,881 people in 2023. This positive trend will continue in 2024 with a total of 44,112 visitors, and increase again to 48,985 visitors in 2025. The increase in the number of visitors indicates that the innovation of digital-based services implemented by the Gowa Regency Library and Archives Office is able to increase public interest in utilizing library services. In addition to being influenced by the availability of digital services, the increase also shows that the public is beginning to

accept service transformation that combines conventional services with information technology.

In addition to increasing the number of direct visits, the successful implementation of the Smart Library Program can also be seen from the development of the number of users of the iGowa application as a digital library service.

Table 6. Number of iGowa Application Users in 2022–2026

No.	Year	Visitors
1.	2022	30 Visitors
2.	2023	59 Visitors
3.	2024	56 Visitors
4.	2025	70 Visitors
5.	2026	28 Visitors

Source: Gowa Regency Library and Archives Service (2025).

Based on Table 4, the number of iGowa application users shows a fairly good development as one of the indicators of the use of Smart Library services. In 2022, the number of users was recorded at 30 people, then increased to 59 people in 2023. Despite a slight decline to 56 people in 2024, the number of users increased again to 70 people in 2025. Meanwhile, data for 2026 shows 28 users, but this figure is provisional data until the time the research is carried out so it cannot be directly compared with the previous annual data.

The data shows that people are starting to use digital library services as an alternative to obtain information without having to come directly to the library. Although the number of iGowa application users is still lower than the overall number of library visitors, this condition shows a change in service utilization patterns towards a more digital system. Therefore, continuous efforts are needed through socialization, increasing people's digital literacy, and developing digital features and collections so that the use of the iGowa application can continue to increase.

The findings were strengthened by the results of interviews with several service users who stated that the existence of the iGowa and OPAC applications provides convenience in finding and accessing library collections. Users no longer have to search for books manually because information regarding the availability of collections can be obtained through digital systems. In addition, digital library services are considered more flexible because they can be accessed at any time according to the needs of users.

The results of the researcher's observations also show that most of the target groups have been able to make good use of the Smart Library facilities, especially among students and students who have a higher level of digital literacy. However, some people still need assistance from library staff in operating the iGowa and OPAC applications. This condition shows that the successful implementation of the Smart Library Program is not only influenced by the availability of technology, but also by the ability of the community to utilize this technology.

Based on the perspective of (Korten, 1987), the implementation of the program can be said to be successful if there is a match between the needs of the target group and the benefits provided by the program. The findings of the study show that the Smart Library Program has provided real benefits to the community through increasing the ease of access to information, service efficiency, and flexibility in utilizing library services. However, increasing people's digital literacy, expanding digital collections, and optimizing the use of the iGowa application still needs to be carried out so that all target groups can benefit maximally from the implementation of the Smart Library Program.

4. Conclusion

The implementation of Smart Library at the Gowa Regency Library and Archives Office based on David C. Korten's theory shows that the implementing element has played a good role in supporting the

implementation of the program. The organizational structure, division of tasks, and coordination between employees have supported the implementation of digital-based library services. However, improving the competence of human resources and strengthening technological infrastructure is still needed to optimize the implementation of the program.

The Smart Library program has been implemented through the provision of the iGowa application, OPAC, the development of library collections, and the use of social media as a means of disseminating information. These various innovations provide convenience for the public to access library services. However, the development of digital collections, improving the quality of the internet network, and improving digital services still need to be carried out so that the implementation of the program is more effective.

From the aspect of the target group, the Smart Library Program has provided benefits for students, students, teachers, civil servants, and the general public. The increase in the number of library visitors and users of the iGowa application shows that the program has been accepted and utilized by the community. Therefore, increasing digital literacy and developing technology-based services needs to continue to be carried out so that the benefits of the Smart Library Program can be felt more widely and sustainably.).

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